

APURV KUSHWAHA

Robotics, Perception and Embedded Systems

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EDUCATION

Master of Science in Robotics 3.83/4.00	Aug 2025 – May 2027
<i>University of Minnesota, Twin Cities</i>	<i>Minneapolis, MN</i>
Bachelor of Technology in Engineering Physics	Jul 2020 – May 2024
<i>Indian Institute of Technology Guwahati</i>	<i>Guwahati, India</i>

EXPERIENCE

Graduate Researcher	Apr 2026 – Present
<i>University of Minnesota, Twin Cities</i>	<i>Minneapolis, MN</i>
- Developing human-in-the-loop robot manipulation systems that combine Meta Quest 3 teleoperation with CAD/RGB-D tracking of a 3D-printed leader arm for a Trossen WidowX AI manipulator. - Implemented task-frame calibration, articulated-joint tracking, inverse kinematics, and safety-gated MuJoCo validation; validated Quest-based hardware control at 90 Hz with zero IK failures in a 51-second run.	
Lead Technical Engineer	Nov 2024 – Jun 2025
<i>Samaritan AI</i>	<i>New Delhi, India</i>
- Led end-to-end development of a smart medical device, covering mechanical design, electronics integration, embedded firmware, and prototype validation. - Designed and fabricated mechanical parts using CAD and 3D printing, assembled electronic subsystems, and integrated sensors into a working prototype presented to 3+ investors and hospitals. - Developed ESP32 firmware for real-time sensing and device operation, and built 24-hour reliability tests that improved signal stability and reduced measurement inconsistency by 87%.	
Research & Development Engineer	May 2024 – Nov 2024
<i>Swachh.io</i>	<i>New Delhi, India</i>
- Supported end-to-end development of environmental monitoring and pollution-control prototypes, contributing to design, testing, and system integration for real-world deployment. - Coordinated 4+ interns across mechanical, electronics, and software domains, and contributed simulation and validation work used in reports that helped secure a \$50K+ grant.	

SELECTED PROJECTS

Quest 3 Mixed-Reality Bowling Analysis Code Demo <i>Unity, YOLO, SAM2, Python</i>	Apr 2026
- Collaborated on a course team system combining Quest 3 H.264 capture at 1280×960 and 30 FPS, timestamped metadata, YOLO-seeded SAM2 tracking, and mixed-reality trajectory replay. - Contributed to validation of a real Quest-to-laptop hotspot session as a decodable H.264 stream and projection of tracked trajectories into lane coordinates for replay and shot analysis.	
Semantic-Aware Image Retargeting Code Demo <i>PyTorch, DINOv3, Depth Anything 3, OpenCV</i>	Dec 2025
- Built a semantic-aware image retargeting pipeline that combines saliency detection, depth estimation, and seam carving to resize images while preserving important content and visual composition. - Trained a DINOv3-based classifier on the PICD dataset (~43K images), achieving repo-reported 75–80% validation accuracy for composition-aware retargeting.	
Autonomous Car Parking Bot Code <i>ROS 2, Gazebo, Nav2, OpenCV</i>	Nov 2025
- Built an autonomous indoor parking system in ROS 2 and Gazebo, combining overhead-camera vehicle detection, Nav2-based navigation, and a towing robot with lift control. - Implemented coordinate transformation and fine-positioning logic for a parking lot, yielding highly precise orientation detection, ±10 cm positioning accuracy, and 100% navigation success in tests.	

SKILLS

Programming & Robotics: Python, C++, Embedded C, MATLAB, ROS 2, Gazebo, Path Planning, SLAM
Perception & AI: OpenCV, Sensor Fusion, Computer Vision, Machine Learning, Reinforcement Learning
Mechatronics: ESP32, Arduino, PCB Design, CAD, Embedded Systems, Sensor Integration, I2C, SPI, UART, CAN
Engineering Tools: Linux, Git, Docker, CMake, SolidWorks, Fusion 360, ANSYS, 3D Printing